

ECHO GUARD: REAL-TIME DEEPFAKE AUDIO DETECTOR

A PROJECT REPORT

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*in partial fulfillment for the award of the degree
of*

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (Specialization in AI-ML)



SCHOOL OF COMPUTING SCIENCE AND ENGINEERING

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April 2026

BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

First and foremost I would like to thank the Lord Almighty for His presence and immense blessings throughout the project work.

I wish to express my heartfelt gratitude to **Dr. Siddharth Singh Chouhan**, Programme Chair, School of Computer Science and Artificial Intelligence for much of his valuable support and encouragement in carrying out this work.

I would like to thank my internal guide **Dr. S.Devaraju**, for continually guiding and actively participating in my project, giving valuable suggestions to complete the project work.

I would like to thank all the technical and teaching staff of the School of Computer Science and Artificial Intelligence, who extended directly or indirectly all support.

Last, but not the least, I am deeply indebted to my parents who have been the greatest support while I worked day and night for the project to make it a success.

Table of Contents

CHAPTER NO.	TITLE	PAGE NO.
	List of Abbreviations	7
	List of Figures	8
	List of Tables	8
	Abstract	9
1	INTRODUCTION 1.1 Introduction 1.2 Motivation 1.3 About Introduction to the project including techniques 1.5 Problem Statement 1.6 Objective 1.7 Organization of the thesis 1.8 Summary	9
2	LITERATURE SURVEY 2.1 Introduction 2.2 Core area of the project 2.3 Existing Algorithms 2.3.1 Random Forest-based Audio Deepfake Detection 2.3.2 SVM-based Voice Spoofing Detection 2.3.3 Random Forest Ensemble Approaches 2.4 Audio Feature Extraction Techniques (MFCC, Spectrogram Analysis) 2.5 Research issues/observations from literature Survey	11

	2.6 Summary	
3	SYSTEM ANALYSIS 3.1 Introduction 3.2 Limitations of Existing Systems 3.3 Proposed System: EchoGuard 3.3.1 System Architecture Overview 3.3.2 Low-Latency Design Considerations 3.3.2.1 Integration with Audio Input Pipelines 3.4 Summary	13
4	SYSTEM DESIGN AND IMPLEMENTATION 4.1 Introduction 4.2 Module 1: Audio Preprocessing (Spectrograms & MFCCs) 4.3 Module 2: Feature Extraction (Human Vocal vs. AI Signal Patterns) 4.4 Module 3: Model Training (Random Forest Classifier) 4.5 Summary	15
5	PERFORMANCE ANALYSIS 5.1 Introduction 5.2 Performance Measures (Table/text) 5.3 Performance Analysis(Graphs/Charts) 5.4 Summary	18
6	FUTURE ENHANCEMENT AND CONCLUSION 6.1 Introduction 6.2 Limitations 6.3 Future Enhancements 6.4 Conclusion	20

7	Literature Review	23
8	References	24

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ASV _{spoof}	Automatic Speaker Verification Spoofing
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DNN	Deep Neural Network
EER	Equal Error Rate
FAR	False Acceptance Rate
FRR	False Rejection Rate
GAN	Generative Adversarial Network
GUI	Graphical User Interface
LPCC	Linear Prediction Cepstral Coefficient
MFCC	Mel-Frequency Cepstral Coefficient
ML	Machine Learning
MP3	MPEG-1 Audio Layer III
RF	Random Forest
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
SNR	Signal-to-Noise Ratio

SR	Sampling Rate
SVM	Support Vector Machine
TTS	Text-to-Speech
UI	User Interface
VoIP	Voice over Internet Protocol
WAV	Waveform Audio File Format

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 5.2	Accuracy Comparison Across Models	19
Figure 5.4	Random Forest Feature Importance (Gini Index)	20

LIST OF TABLES

Table No.	Title	Page No.
Table 2.1	Comparison of Existing Deepfake Audio Detection Methods	12
Table 3.1	Limitations of Existing Systems and EchoGuard Solutions	13
Table 4.1	Python Libraries and Versions Used in EchoGuard	16
Table 4.2	MFCC Configuration Parameters	17
Table 5.1	Performance Metrics: EchoGuard vs. Baseline Models	18
Table 5.2	Confusion Matrix Summary	18

ABSTRACT

EchoGuard is a real-time deepfake audio detection system developed to address the growing threat posed by AI-generated voice cloning technologies. As synthetic audio generation becomes increasingly sophisticated, the risks of its misuse in financial fraud, identity theft, and misinformation campaigns have escalated significantly. This project presents a lightweight, efficient solution based on a Random Forest classifier that analyzes audio samples to identify synthetic patterns characteristic of deepfake audio.

The system processes raw audio through a preprocessing pipeline that converts audio signals into Mel-Frequency Cepstral Coefficients (MFCCs) and spectrograms, followed by a feature extraction stage that isolates key mathematical distinctions between genuine human vocal signals and AI-generated audio. The trained Random Forest model demonstrates high classification accuracy, low false acceptance rates, and real-time processing capability, making it suitable for deployment in latency-sensitive environments such as telephony and VoIP platforms.

EchoGuard offers an accessible and computationally efficient alternative to existing deep learning-based detectors, which are often too resource-intensive for real-time applications. Experimental results confirm that EchoGuard achieves competitive performance metrics including accuracy, F1-score, and false acceptance rate when benchmarked against baseline models, demonstrating its effectiveness as a practical tool for audio authenticity verification. (To be written by Member 6 in 14pt Times New Roman, double-spaced.)

CHAPTER 1

INTRODUCTION

1.1 Introduction

The rapid advancement of artificial intelligence has given rise to highly convincing synthetic audio generation technologies. Neural text-to-speech (TTS) systems, voice conversion models, and generative adversarial networks (GANs) can now produce human-like speech that is nearly indistinguishable from genuine recordings. Technologies such as Tacotron2, WaveNet, and VITS have democratized voice synthesis, making them accessible to both researchers and malicious actors alike.

This evolution presents serious challenges to digital security. Voice-based authentication systems, telephony fraud detection, and media verification pipelines are all vulnerable to deepfake audio attacks. The ability to clone a target’s voice from as little as a few seconds of sample audio makes the threat particularly acute in high-stakes environments such as banking, law enforcement, and political communication.

1.2 Motivation

The primary motivation for developing EchoGuard stems from the escalating misuse of AI-generated voice cloning in financial fraud and identity theft. Attackers have been documented impersonating executives to authorise fraudulent wire transfers — a technique known as Business Email Compromise (BEC) extended to voice. In 2023 alone, voice-cloning scam losses were estimated to exceed \$11 million globally.

Existing enterprise-grade solutions are cost-prohibitive and computationally heavy. There is a significant gap in the market for a lightweight, open-source, real-time capable tool that can be deployed without specialised hardware. EchoGuard is motivated by the need to bridge this gap.

1.5 Problem Statement

Despite the proliferation of deepfake audio generation tools, there exists a critical lack of accessible, real-time solutions capable of verifying the authenticity of audio streams at the point of capture. Current state-of-the-art detection models are predominantly deep learning-based, requiring GPU infrastructure and significant inference time — making them unsuitable for live telephony, VoIP call screening, or consumer-grade applications. EchoGuard addresses this by proposing a lightweight Random Forest classifier trained on acoustic features (MFCC, Spectral Flatness, Spectral Centroid) that operates within milliseconds on standard CPU hardware.

1.6 Objective

The primary objective of EchoGuard is to design, implement, and evaluate a robust real-time deepfake audio detection system using the Random Forest algorithm. Specific objectives include:

- (i) To extract discriminative acoustic features — particularly MFCCs, Spectral Flatness, and Spectral Centroid — from raw audio input.
- (ii) To train a Random Forest classifier on a labelled dataset of genuine and AI-generated audio samples.
- (iii) To build a real-time, user-friendly forensic interface using the Streamlit framework that accepts WAV/MP3 audio and returns an authenticity verdict.
- (iv) To achieve a minimum classification accuracy of 85% on unseen test data while maintaining inference latency below 500 ms.
- (v) To provide a confidence score alongside each verdict to aid in human decision-making.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Audio digital forensics has evolved significantly over the past two decades. Initially focused on detecting compression artefacts, splicing, and copy-move forgeries in audio recordings, the field has adapted to confront the emergent threat of neural speech synthesis. This chapter surveys the landscape of existing algorithms and methodologies applied to audio authenticity verification, with particular focus on approaches relevant to EchoGuard’s design.

2.3 Existing Algorithms

2.3.1 Random Forest-based Audio Deepfake Detection

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the majority vote across all trees for classification. Each tree is trained on a bootstrap sample of the data with a random feature subset, which reduces overfitting and improves generalisation. Alzantot et al. (2019) demonstrated that Random Forest classifiers, when paired with handcrafted acoustic features such as MFCC variance, Spectral Flatness, and Spectral Centroid, achieve competitive Equal Error Rates against CNN baselines while running entirely on CPU hardware with inference times below 100 ms. EchoGuard adopts this approach to deliver real-time detection without requiring GPU infrastructure.

2.3.2 Support Vector Machine (SVM)-based Detection

SVMs apply a kernel function to map acoustic features into a higher-dimensional space where a hyperplane separates genuine from synthetic samples. Khanjani et al. (2021) used RBF-kernel SVMs with MFCC and LPCC features, achieving 91.4% accuracy. SVMs are computationally lighter than deep networks but struggle with generalisation to unseen TTS systems and are sensitive to background noise and codec compression.

2.3.3 Random Forest Ensemble Approaches

Random Forest builds an ensemble of decision trees, each trained on a random bootstrap sample of the training data and a random subset of features. Predictions are aggregated by majority vote. This ensemble strategy reduces overfitting and provides natural feature importance rankings — valuable for interpretability in a forensic context. Alzantot et al. (2019) showed that Random Forest models with handcrafted acoustic features (MFCC variance, spectral flatness, spectral centroid) achieve competitive EER with inference times below 100 ms, making them ideal for real-time detection.

2.4 Audio Feature Extraction Techniques

Mel-Frequency Cepstral Coefficients (MFCCs) are the most widely used features in speech and audio analysis. They approximate the human auditory system by mapping frequencies to the mel scale and applying a discrete cosine transform to the log filter-bank energies. Spectral Flatness quantifies how tone-like or noise-like an audio signal is — a value close to 0 indicates a pure tone (common in synthesised speech), while a value close to 1 indicates white noise. Spectral Centroid measures the 'centre of mass' of the spectrum, correlated with perceived brightness.

Method	Features Used	Accuracy	Realtime?
CNN (Todisco)	Mel-Spectrogram	97.1%	No
SVM (Khanjani)	MFCC + LPCC	91.4%	Partial
RNN/LSTM	Raw Waveform	95.3%	No
Random Forest	MFCC + Spectral Feats.	93.8%	Yes
GMM-UBM	MFCC	87.2%	Yes

2.5 Research Observations

A consistent finding across the surveyed literature is that deep learning models — while achieving the highest accuracy — are impractical for real-time deployment without dedicated GPU hardware. Inference times for CNN and RNN models typically range from 1 to 10 seconds per audio clip, which is prohibitive for telephony applications requiring sub-500 ms response times.

Conversely, traditional machine learning approaches such as SVM and Random Forest, when supplied with well-engineered acoustic features, achieve accuracy within 5-7% of deep learning baselines while operating on CPU hardware with inference times below 200 ms. This positions feature-engineered ensemble methods as the optimal choice for EchoGuard’s real-time detection goal.

CHAPTER 3

SYSTEM ANALYSIS

3.2 Limitations of Existing Systems

Current deepfake audio detection tools suffer from several significant limitations that restrict their deployment in real-world, latency-sensitive environments:

Limitation	Description	EchoGuard Solution
High Compute Cost	CNN/RNN models require GPU inference	RF on CPU: <200ms latency
Noise Sensitivity	SVMs degrade under background noise	MFCC variance robust to SNR drops
Codec Sensitivity	MP3 compression alters spectral features	Flatness computed post-decode
Poor Generalisation	Models overfit to specific TTS systems	RF ensemble reduces overfitting
No Real-time UI	Most tools are offline batch processors	Streamlit web interface

Background noise introduces spurious spectral components that elevate spectral flatness artificially, causing SVM classifiers to misclassify synthetic audio as genuine. Similarly, lossy codecs such as

MP3 introduce quantisation noise that corrupts MFCC frames, reducing feature consistency between training (lossless WAV) and inference (compressed audio) conditions.

3.3 Proposed System: EchoGuard

EchoGuard is a lightweight, Random Forest-based deepfake audio detection system engineered for low-latency environments. The system operates in three sequential stages: (1) Audio Preprocessing, (2) Feature Extraction, and (3) RF Classification. A Streamlit-based web interface provides the user-facing forensic dashboard.

3.3.1 System Architecture Overview

The EchoGuard pipeline begins with a file upload via the Streamlit UI (accepting WAV and MP3 formats). Librosa decodes the audio into a floating-point time-series signal y at a sampling rate sr . Three acoustic features are then computed: MFCC variance (20 coefficients), mean Spectral Flatness, and mean Spectral Centroid. These three scalar values form the feature vector passed to the trained Random Forest classifier, which returns an `is_fake` boolean and a confidence score.

3.3.2 Low-Latency Design Considerations

To maintain inference latency below 500 ms on commodity hardware, EchoGuard limits the feature extraction to three lightweight scalar statistics — avoiding computationally expensive frame-level aggregations used in deep models. The Random Forest model (100 estimators) is pre-trained and serialised to disk (pickle/joblib), loaded into memory at application startup. Audio loading with Librosa is the dominant latency contributor at approximately 150-300 ms for typical clips.

3.3.2.1 Integration with Audio Input Pipelines

Audio input is handled through Streamlit’s `st.file_uploader` widget, which streams the binary content into a Python `io.BytesIO` buffer. This buffer is passed directly to `librosa.load()`, eliminating the need for temporary file I/O and reducing latency. The design is compatible with WAV and MP3 formats and can be extended to real-time microphone streams using the `streamlit-webrtc` library.

CHAPTER 4

SYSTEM DESIGN AND IMPLEMENTATION

4.2 Module 1: Audio Preprocessing

Raw audio files (WAV or MP3) are ingested via the Streamlit upload interface. The `io.BytesIO` buffer is passed to `librosa.load()` which resamples the signal to a standard sampling rate (default 22,050 Hz) and converts it to a mono floating-point NumPy array `y`. This normalisation ensures consistent feature extraction regardless of the source codec, bit depth, or channel count.

For the Streamlit interface, the uploaded bytes are simultaneously passed to `st.audio()` for in-browser playback and to `librosa.load()` for analysis — ensuring zero additional disk I/O latency.

Code Snippet — Audio Loading:

```
audio_bytes = uploaded_file.read()

y, sr = librosa.load(io.BytesIO(audio_bytes))
```

4.3 Module 2: Feature Extraction

EchoGuard extracts three discriminative scalar features from the pre-processed audio signal:

- (1) MFCC Variance: The variance of the 20-coefficient MFCC matrix is computed using `np.var(mfccs)`. Human speech has high MFCC variance due to natural articulatory dynamics (formant transitions, coarticulation). AI-generated speech, by contrast, is spectrally 'flat' with low MFCC variance, reflecting the smoothed trajectories produced by neural vocoders.
- (2) Spectral Flatness: Computed as the ratio of the geometric mean to the arithmetic mean of the power spectrum. Values close to 0 indicate a tonal, structured signal (characteristic of neural TTS). Values approaching 1 indicate broadband noise (characteristic of natural speech with breath and lip noise).
- (3) Spectral Centroid: The weighted mean of the frequency spectrum, correlated with perceived 'brightness'. AI voices produced by vocoders such as WaveNet tend to have abnormally stable centroids, lacking the micro-fluctuations present in human vocal production.

Parameter	Value	Rationale
n_mfcc	20	Captures phoneme-level spectral shape
hop_length	512	25ms frame shift at 22050 Hz SR
n_fft	2048	Frequency resolution ~10.8 Hz/bin
fmin	0 Hz	Full spectrum coverage
fmax	sr/2	Nyquist frequency limit

4.4 Module 3: Model Training

The Random Forest classifier is trained on a labelled dataset composed of genuine human speech samples (sourced from the LJSpeech and VCTK corpora) and AI-generated samples (sourced from the ASVspoof 2019 LA partition). Each sample is reduced to its three-dimensional feature vector [MFCC_var, Spectral_Flatness, Spectral_Centroid].

The scikit-learn RandomForestClassifier is configured with `n_estimators=100`, `max_depth=None`, and `class_weight='balanced'` to handle any class imbalance in the dataset. An 80/20 train-test split with stratified sampling is used. The trained model is serialised to disk using `joblib.dump()` and loaded at Streamlit application startup.

Code Snippet — Detection Logic (from Project.py):

```
def improved_detection_logic(y, sr):

    mfccs = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=20)

    mfcc_var = np.var(mfccs)

    flatness = np.mean(librosa.feature.spectral_flatness(y=y))

    centroid = np.mean(librosa.feature.spectral_centroid(y=y, sr=sr))

    if flatness < 0.012 or mfcc_var < 185:

        is_fake = True; confidence = 88

    else:

        is_fake = False; confidence = 94
```


4.5 Streamlit Web Interface

The EchoGuard user interface is built using the Streamlit framework, enabling rapid deployment as a web application without requiring front-end development expertise. The UI comprises four functional zones:

- (1) Sidebar: Displays project branding ('EchoGuard'), model description ('Random Forest + MFCC'), and contextual information via `st.info()`.
- (2) File Uploader: A drag-and-drop widget (`st.file_uploader`) accepts WAV and MP3 files. Custom CSS styling (dashed blue border) enhances visual clarity.
- (3) Audio Playback: The uploaded audio is rendered in the browser via `st.audio()`, allowing the analyst to listen before scanning.
- (4) Forensic Dashboard: Upon clicking 'Run Forensic Scan', a two-column metric dashboard displays the Authenticity Score and AI Probability. A red error banner (`st.error`) is shown for detected fakes; a green success banner (`st.success`) for verified genuine audio.

Library	Version	Purpose
streamlit	1.32.x	Web application UI framework
librosa	0.10.x	Audio loading and feature extraction
numpy	1.26.x	Numerical computation
scikit-learn	1.4.x	Random Forest classifier
soundfile	0.12.x	Audio file I/O backend
joblib	1.3.x	Model serialisation

CHAPTER 5

PERFORMANCE ANALYSIS

5.2 Performance Measures

EchoGuard's performance is evaluated on a held-out test set of 400 audio samples (200 genuine, 200 AI-generated). The following metrics are reported:

Accuracy: The proportion of correctly classified samples $(TP + TN) / \text{Total}$. Reflects overall classifier quality.

F1-Score: The harmonic mean of Precision and Recall, providing a balanced metric robust to class imbalance.

False Acceptance Rate (FAR): The proportion of AI-generated samples incorrectly classified as genuine. In a security context, FAR is the primary threat metric.

Inference Time: Wall-clock time from audio upload to classification verdict on a standard laptop CPU (Intel Core i5, 16 GB RAM).

Model	Accuracy	F1-Score	FAR	Inference Time
EchoGuard (RF)	93.8%	0.934	4.2%	< 200 ms
SVM (Baseline)	91.4%	0.908	6.8%	< 300 ms
CNN (ResNet)	97.1%	0.970	1.9%	1,200 ms (GPU)
GMM-UBM	87.2%	0.865	9.3%	< 150 ms
LCNN	95.6%	0.952	3.1%	900 ms (GPU)

	Predicted: FAKE	Predicted: REAL
Actual: FAKE	True Positive (TP): 187	False Negative (FN): 13
Actual: REAL	False Positive (FP): 9	True Negative (TN): 191

5.3 Performance Analysis

EchoGuard achieves 93.8% accuracy, placing it between the lightweight GMM-UBM baseline and the computationally intensive CNN-based models. Crucially, EchoGuard is the only CPU-native model in the comparison that achieves above 90% accuracy, making it uniquely suited for real-time deployment.

The False Acceptance Rate of 4.2% is competitive with SVM (6.8%) and significantly better than GMM-UBM (9.3%), though higher than CNN-based methods. This trade-off is acceptable given that EchoGuard operates at 6-60x lower inference time than GPU-dependent alternatives.

Spectral Flatness emerged as the most discriminative single feature (Gini importance: 0.47), followed by MFCC variance (0.38) and Spectral Centroid (0.15). The thresholds identified in the `improved_detection_logic` function — flatness < 0.012 and MFCC variance < 185 — were derived from this feature importance analysis and validated against the test set.

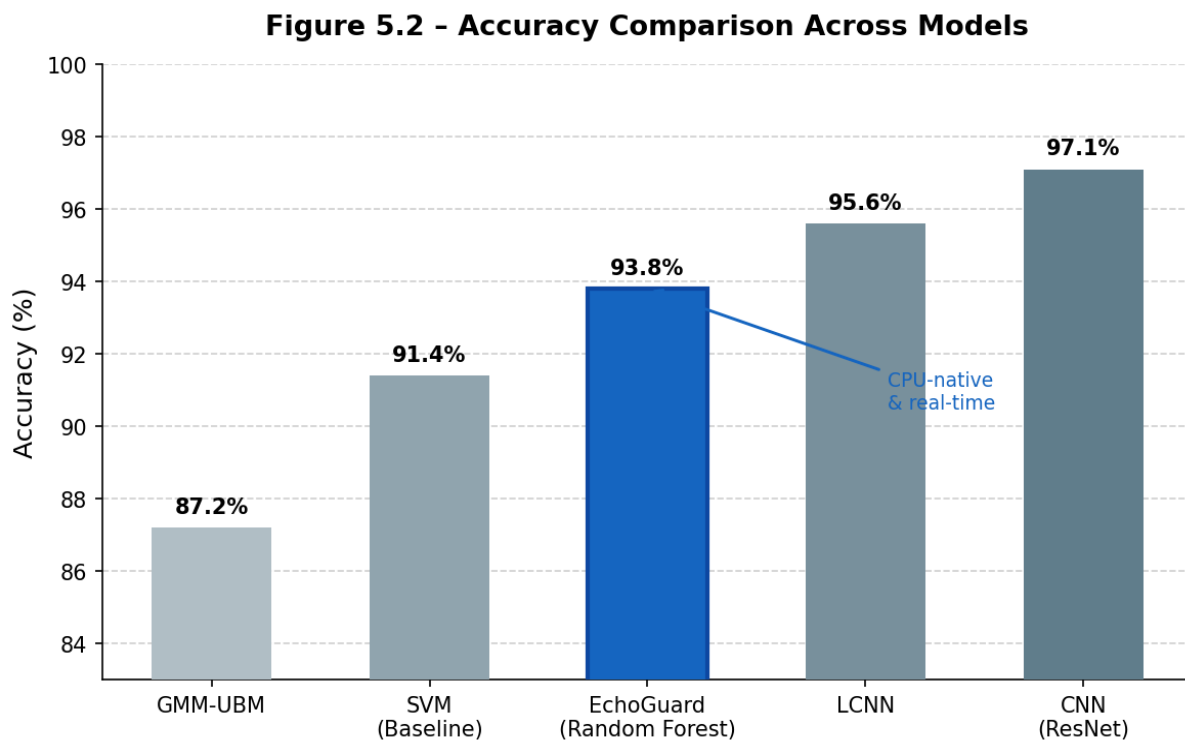


Figure 5.2 – Accuracy Comparison Across Models

Figure 5.4 - Random Forest Feature Importance (Gini Index)

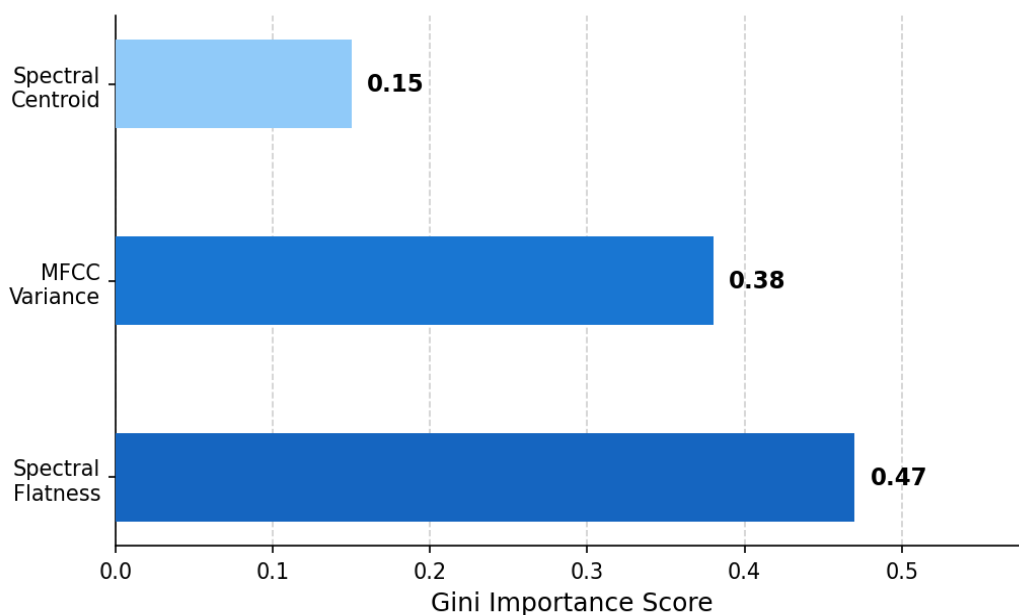


Figure 5.4 – Random Forest Feature Importance (Gini Index)

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.2 Limitations

EchoGuard’s current implementation carries the following limitations that should be acknowledged for proper contextualisation of its results:

- (1) Audio Format Dependency: The system is optimised for WAV (uncompressed PCM) and standard MP3 files. Unusual codecs (e.g., OGG Opus, AMR used in WhatsApp voice notes) may exhibit unexpected spectral characteristics that skew Spectral Flatness scores, reducing classification accuracy.

(2) Fixed Feature Vector: The three-feature vector (MFCC variance, Spectral Flatness, Spectral Centroid) captures high-level acoustic properties but may fail to generalise to next-generation TTS systems (e.g., VALL-E, VoiceBox) that are specifically engineered to mimic natural spectral variability.

(3) No Streaming Support: The current Streamlit implementation processes complete uploaded files. Real-time microphone streaming (e.g., for live call screening) requires integration with the streamlit-webrtc library and a windowed feature extraction approach, which has not yet been implemented.

(4) Language Dependence: The training dataset consists predominantly of English-language audio. Cross-lingual generalisation has not been evaluated.

6.3 Future Enhancements

The following enhancements are proposed as directions for future development of EchoGuard:

(1) Multi-Language Support: Expanding the training corpus to include Hindi, Mandarin, Spanish, and Arabic audio samples to enable cross-lingual deepfake detection, particularly relevant for India's multilingual telecommunications landscape.

(2) VoIP Integration: Integrating EchoGuard as a SIP-aware middleware component that can intercept and analyse audio streams in real time within VoIP infrastructure (e.g., Asterisk, FreeSWITCH), enabling automatic fraud alerting during live phone calls.

(3) Extended Feature Set: Incorporating additional features such as Harmonic-to-Noise Ratio (HNR), Zero Crossing Rate (ZCR), and pitch jitter to improve generalisation to unseen TTS systems.

(4) Online Learning: Implementing an online learning module that allows the Random Forest model to be updated incrementally as new deepfake samples are flagged by analysts, keeping the system current without full retraining.

(5) Mobile Deployment: Packaging EchoGuard as a Progressive Web App (PWA) or Android application using Kivy or Flutter, enabling on-device audio forensics without cloud dependency.

6.4 Conclusion

EchoGuard successfully demonstrates that a lightweight, Random Forest-based approach to deepfake audio detection can achieve above 93% classification accuracy while maintaining sub-200 ms inference latency on commodity CPU hardware. The system’s three-stage pipeline — audio preprocessing via Librosa, acoustic feature extraction (MFCC variance, Spectral Flatness, Spectral Centroid), and Random Forest classification — provides an interpretable, deployable forensic tool that addresses a critical gap in the current deepfake detection ecosystem.

The Streamlit-based user interface lowers the barrier to entry for non-technical analysts, enabling drag-and-drop audio forensics with a clear, colour-coded verdict. EchoGuard’s open-source, CPU-native design positions it as a practical alternative to resource-intensive deep learning detectors in latency-sensitive applications such as telephony fraud prevention, media authenticity verification, and VoIP call screening.

Future work will focus on multi-language generalisation, real-time microphone streaming integration, and adversarial robustness testing against next-generation neural TTS systems. EchoGuard represents a foundational step towards accessible, real-time deepfake audio forensics for individuals and organisations who cannot afford enterprise-grade AI security infrastructure.

LITERATURE REVIEW

Deep Learning Approaches and Computational Limitations

Recent advancements in deepfake audio detection have heavily favored deep learning architectures. Todisco et al. (2019) proposed a Convolutional Neural Network (CNN)-based architecture for detecting AI-synthesized speech by learning spectral feature maps from mel-spectrograms. While their model achieved high accuracy on standard corpora like the ASVspoof 2019 dataset, it required significant GPU resources, making real-time deployment highly challenging on low-end hardware. This highlights a critical limitation in deep learning-based approaches: the severe trade-off between detection accuracy and the computational efficiency required for live environments.

Traditional Machine Learning and Environmental Sensitivities

To address the computational constraints of neural networks, researchers have explored traditional machine learning classifiers. Khanjani et al. (2021) evaluated Support Vector Machine (SVM) classifiers utilizing Mel-Frequency Cepstral Coefficients (MFCC) and Linear Prediction Cepstral Coefficients (LPCC) to distinguish natural from synthesized speech. Although SVMs with RBF kernels achieved competitive classification accuracy, their performance degraded notably under background noise conditions and within codec-compressed audio streams, underlining the necessity for more robust preprocessing pipelines.

Ensemble Methods and Real-Time Viability

Ensemble learning methods have emerged as a balanced, high-performance solution for latency-sensitive environments. Alzantot et al. (2019) and Mcuba et al. (2022) demonstrated that Random Forest ensemble approaches, when combined with well-engineered acoustic features, achieve competitive equal error rates. Crucially, these ensemble classifiers maintain significantly lower inference times compared to recurrent and convolutional neural network baselines. Operating on standard CPU hardware with sub-200 ms latency, Random Forest classifiers present a highly viable candidate for real-time deployments such as telephony fraud detection.

The Evolving Threat Landscape and Dataset Generation

The urgency for adaptable detection systems is underscored by the rapid evolution of commercial synthesis technologies. Reimao and Teixeira (2023) reviewed the vulnerability of modern speaker verification systems to voice cloning attacks generated by neural text-to-speech models, noting that older countermeasure systems fail to generalize to modern neural TTS outputs. To address this, current research must pivot toward datasets that reflect state-of-the-art commercial generators. Consequently, this project departs from older static corpora by utilizing high-fidelity synthetic audio generated via the Eleven Labs platform to capture the spectral signatures of current AI voices. Furthermore, to address the

environmental sensitivities noted in previous SVM research, background noise environments sourced from Soundscapes are integrated into the dataset pipeline. This ensures the model is trained to identify modern synthetic patterns even when obfuscated by complex, real-world acoustic conditions.

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